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Course Code: CSA0820

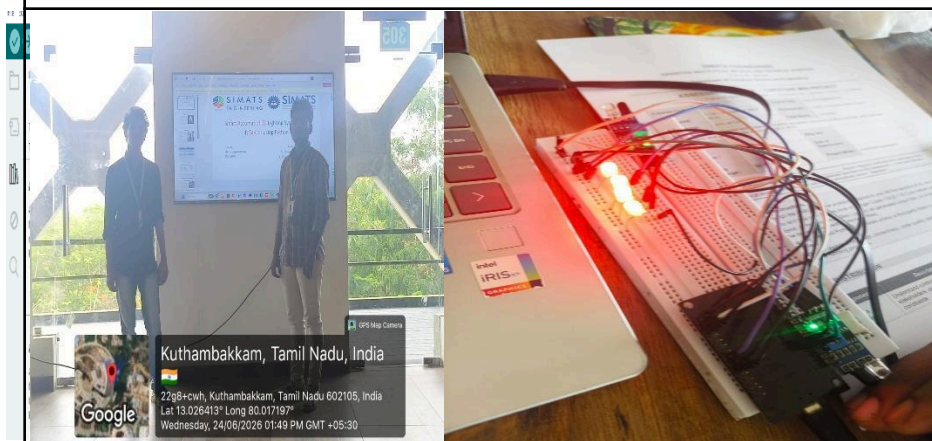
Slot: C

Course Name: Python Programming for Scientific Computing

Course Faculty: Dr . Vijayaragavan

Project Title: Smart Automatic LED Lighting System And IR Sensor using Python

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description.

1. Sensor Data Acquisition

The IR sensor detects the presence or absence of an object (like a person). The microcontroller reads this data as a digital signal (HIGH/LOW) through input pins.

2. Decision-Making Logic

The embedded program processes the sensor input:

- If an object is detected → LED turns ON
- If no object is detected → LED turns OFF

This logic is implemented using conditional statements (if-else) in the code.

3. Microcontroller Programming

The system is programmed using platforms like **Arduino IDE**, where the code is written, compiled, and uploaded to the microcontroller (e.g., Arduino Uno). The software ensures real-time response and efficient execution.

4. Output Control

Based on processed input, the microcontroller sends signals to control the LED via output pins. This ensures automatic lighting without manual intervention.